Differential Geometry - Lecture 1
Differentiable Manifolds

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19 October 2026

About these notes: Gadi Bessudo took the handwritten notes during the lecture, listening to the lecturer and copying from the board. Claude Opus 5.5 (Anthropic) then transcribed them into LyX, redrew the figures, and filled in a few explanations that were missed in class. Errors may come from either step.

1 Motivation: curves and surfaces in $\mathbb{R}^3$

1.1 Curves

A (parametrized) curve in $\mathbb{R}^3$ is a differentiable map $\alpha : I \to \mathbb{R}^3$, where $I \subset \mathbb{R}$ is an open interval. Locally, a curve looks like a piece of $\mathbb{R}$.

A closed curve C, which has the shape of the circle S^1, cannot be described by a single such map that is a homeomorphism onto C (S^1 is compact, an open interval is not). Instead we cover C by (at least) two parametrized pieces $\alpha_1(I_1)$ and $\alpha_2(I_2)$ which overlap (Figure 1).

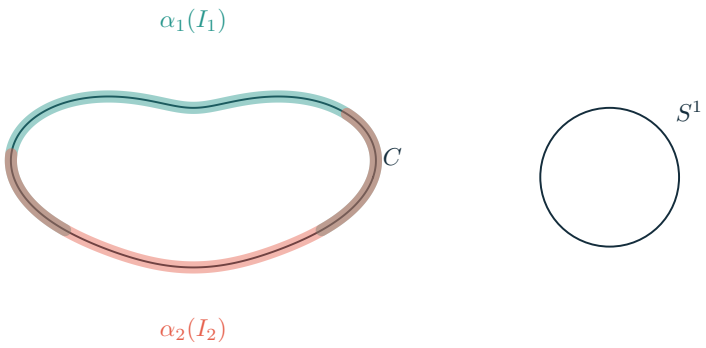


Figure 1: A closed curve is covered by two overlapping parametrized arcs. Topologically it is a circle S^1.

This is the main idea of the course: describe an object by overlapping pieces, each of which looks like an open piece of $\mathbb{R}^n$.

1.2 Regular surfaces

Definition 1. A subset $S \subset \mathbb{R}^3$ is a *regular surface* if for every point $p \in S$ there exist a neighborhood $V \subset \mathbb{R}^3$ of p and a map $\mathbf{x} : U \to V \cap S$ of an open set $U \subset \mathbb{R}^2$ onto $V \cap S$ such that

(a) $\mathbf{x}$ is differentiable and a homeomorphism (it has a continuous inverse $\mathbf{x}^{-1} : V \cap S \to U$);

(b) (regularity) the differential $(d\mathbf{x})_q : \mathbb{R}^2 \to \mathbb{R}^3$ is injective for all $q \in U$.

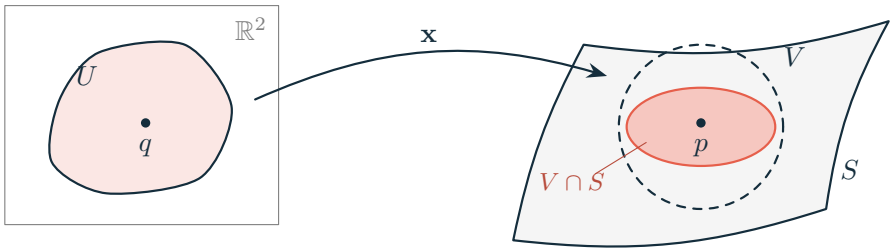


Figure 2: A parametrization $\mathbf{x}$ maps an open set $U \subset \mathbb{R}^2$ onto a neighborhood $V \cap S$ of p in S.

The map $\mathbf{x}$ is called a *parametrization* of S at p, and $V \cap S$ a *coordinate neighborhood* of p.

Intuition: (a) says that $V \cap S$ is just a deformed open piece of the plane, with nothing torn or glued. (b) says that the two partial derivatives $\mathbf{x}_u, \mathbf{x}_v$ are linearly independent, so S has a tangent plane at every point (no cusps or edges). Note that "$\mathbf{x}$ is differentiable" makes sense only because $\mathbf{x}$ takes values in the ambient space $\mathbb{R}^3$.

Example 2. The following are regular surfaces:

- the sphere $S^2 = \{x^2 + y^2 + z^2 = 1\} \subset \mathbb{R}^3$ (covered for example by six graphs $\mathbf{x}(u, v) = (u, v, \pm\sqrt{1 - u^2 - v^2})$ over the open unit disk, and the analogous ones with the roles of x, y, z exchanged);
- the plane $\{z = 0\}$, with the single parametrization $\mathbf{x}(u, v) = (u, v, 0)$;
- every plane $\{ax + by + cz + d = 0\} \subset \mathbb{R}^3$ with $(a, b, c) \neq (0, 0, 0)$.

Remark 3 (Change of parameters). Let $\mathbf{x}_\alpha : U_\alpha \to S$ and $\mathbf{x}_\beta : U_\beta \to S$ be two parametrizations with $W = \mathbf{x}_\alpha(U_\alpha) \cap \mathbf{x}_\beta(U_\beta) \neq \emptyset$. Then the changes of parameters

$$\mathbf{x}_\alpha^{-1} \circ \mathbf{x}_\beta : \mathbf{x}_\beta^{-1}(W) \to \mathbf{x}_\alpha^{-1}(W) \subset \mathbb{R}^2$$

$$\mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha : \mathbf{x}_\alpha^{-1}(W) \to \mathbf{x}_\beta^{-1}(W) \subset \mathbb{R}^2$$

are differentiable (Figure 3).

For a regular surface this is a theorem, and its proof uses the ambient $\mathbb{R}^3$. For an abstract set M there is no ambient space, so "$\mathbf{x}_\alpha$ is differentiable" has no meaning. The only maps we can still ask to be differentiable are the changes of parameters, since they go between open subsets of $\mathbb{R}^n$. This property becomes an axiom in the next definition.

2 Differentiable manifolds in n dimensions

Definition 4. A *differentiable manifold* of dimension n is a set M together with a family of injective mappings $\mathbf{x}_\alpha : U_\alpha \to M$ of open sets $U_\alpha \subset \mathbb{R}^n$ into M such that

(1) $\bigcup_\alpha \mathbf{x}_\alpha(U_\alpha) = M$;

(2) for all α, β with $W = \mathbf{x}_\alpha(U_\alpha) \cap \mathbf{x}_\beta(U_\beta) \neq \emptyset$, the sets $\mathbf{x}_\alpha^{-1}(W)$ and $\mathbf{x}_\beta^{-1}(W)$ are open subsets of $\mathbb{R}^n$, and $\mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha$ and $\mathbf{x}_\alpha^{-1} \circ \mathbf{x}_\beta$ are differentiable functions;

(3) the family $\{(U_\alpha, \mathbf{x}_\alpha)\}$ is maximal relative to conditions (1) and (2).

The pair $(U_\alpha, \mathbf{x}_\alpha)$ with $p \in \mathbf{x}_\alpha(U_\alpha)$ is called a *parametrization* (or system of coordinates) of M at p. A family $\{(U_\alpha, \mathbf{x}_\alpha)\}$ satisfying (1) and (2) is called a *differentiable structure* on M.

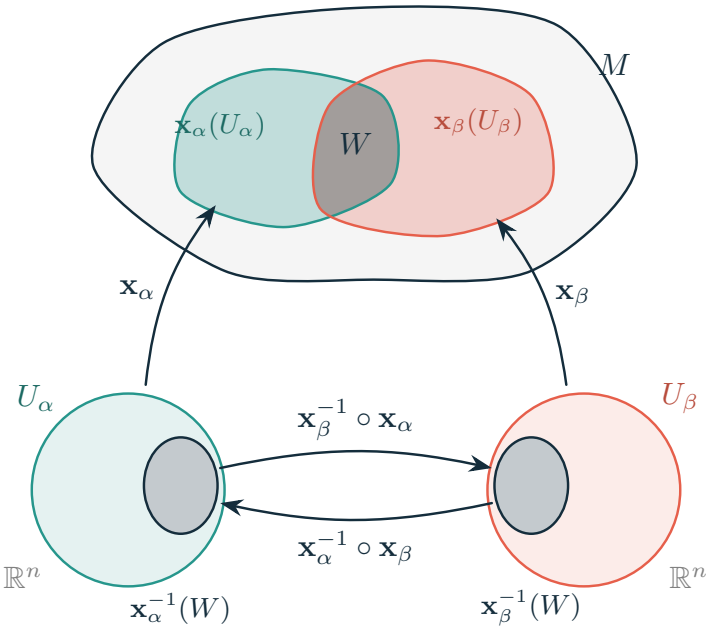


Figure 3: Two overlapping parametrizations. The changes of parameters $\mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha$ and $\mathbf{x}_\alpha^{-1} \circ \mathbf{x}_\beta$ are maps between open subsets of $\mathbb{R}^n$.

Here "differentiable" means C^∞. Condition (1) says the parametrizations cover M, and condition (2) is the compatibility of Remark 3, now imposed as an axiom.

Remark 5 (On maximality). Condition (3) is about the family of parametrizations, not about the size of the sets U_α: any parametrization $\mathbf{y} : U \to M$ that satisfies (2) together with every member of the family is already in the family. It is purely technical: a family satisfying (1) and (2) can always be completed to a maximal one by adding all parametrizations compatible with it. So in practice we define a manifold by giving a differentiable structure, as in the examples below.

Remark 6 (Topology on M). M is only a set, no topology is given. A differentiable structure induces one: declare $A \subset M$ to be *open* if and only if

$$\mathbf{x}_\alpha^{-1}\left(A \cap \mathbf{x}_\alpha(U_\alpha)\right) \text{ is open in } \mathbb{R}^n \text{ for all } \alpha$$

With this topology the sets $\mathbf{x}_\alpha(U_\alpha)$ are open and the maps $\mathbf{x}_\alpha$ are continuous.

3 Example: the real projective space

Definition 7. The real projective space of dimension n is

$$\mathbb{R}P^n := \{\text{lines through the origin in } \mathbb{R}^{n+1}\} = \left(\mathbb{R}^{n+1} \setminus \{0\}\right) / \sim$$

where $u \sim \lambda u$ for all $\lambda \in \mathbb{R}^* = \mathbb{R} \setminus \{0\}$. We write $[x_0, \ldots, x_n]$ for the class of $(x_0, \ldots, x_n)$.

Parametrizations. If $x_0 \neq 0$ we can choose $\lambda = 1/x_0$:

$$[x_0, x_1, \ldots, x_n] = \left[1, \frac{x_1}{x_0}, \ldots, \frac{x_n}{x_0}\right]$$

so such a point is determined by n real numbers. Define

$$V_0 := \{[x_0, \ldots, x_n] \mid x_0 \neq 0\}$$

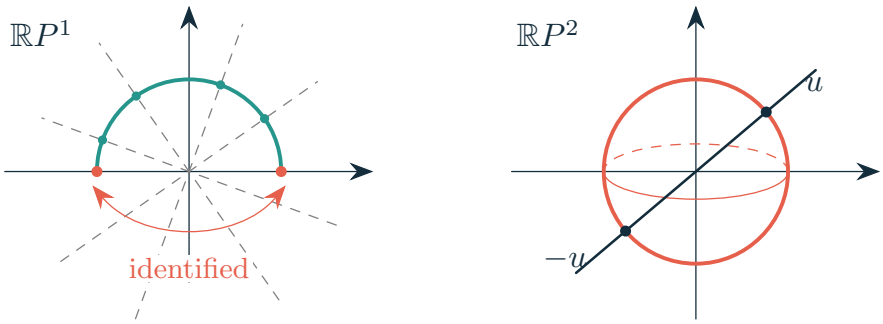


Figure 4: Left: every line through the origin of $\mathbb{R}^2$ meets the upper half circle once, except the horizontal line, which meets both endpoints. Gluing the endpoints gives $\mathbb{R}P^1 \cong S^1$. Right: every line through the origin of $\mathbb{R}^3$ meets S^2 in two antipodal points $\pm u$, so $\mathbb{R}P^2 = S^2/(u \sim -u)$.

and $\mathbf{x}_0 : \mathbb{R}^n \to V_0$ by

$$\mathbf{x}_0(x_1, \ldots, x_n) = [1, x_1, \ldots, x_n]$$

which is injective (in fact a bijection onto V_0). In the same way, for $i = 0, \ldots, n$, let

$$V_i := \{[x_0, \ldots, x_n] \mid x_i \neq 0\}$$

and $\mathbf{x}_i : \mathbb{R}^n \to V_i$ (insert a 1 in slot i)

$$\mathbf{x}_i(y_0, \ldots, y_{i-1}, y_{i+1}, \ldots, y_n) = [y_0, \ldots, y_{i-1}, 1, y_{i+1}, \ldots, y_n]$$

with inverse

$$\mathbf{x}_i^{-1} : [x_0, \ldots, x_n] \mapsto \left(\frac{x_0}{x_i}, \ldots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_n}{x_i}\right)$$

which does not depend on the representative (multiplying all x_k by λ does not change the ratios).

Condition (1). Every $u \neq 0$ has some nonzero coordinate, so

$$\bigcup_{i=0}^{n} V_i = \mathbb{R}P^n$$

Condition (2). Let $i \neq j$. A point $\mathbf{x}_j(y)$ lies in V_i iff its i-th entry is nonzero, so

$$\mathbf{x}_j^{-1}(V_i \cap V_j) = \{(y_0, \ldots, y_{j-1}, y_{j+1}, \ldots, y_n) \in \mathbb{R}^n \mid y_i \neq 0\}$$

is open in $\mathbb{R}^n$ (the entry y_j is omitted). On this set

$$\mathbf{x}_i^{-1} \circ \mathbf{x}_j(y_0, \ldots, y_{j-1}, y_{j+1}, \ldots, y_n) = \left(\frac{y_0}{y_i}, \ldots, \frac{y_{j-1}}{y_i}, \frac{1}{y_i}, \frac{y_{j+1}}{y_i}, \ldots, \frac{y_n}{y_i}\right)$$

where the entry $1/y_i$ sits in slot j and the entry $y_i/y_i = 1$ (slot i) is omitted. All components are rational functions with nonvanishing denominator, hence differentiable.

So $\{(\mathbb{R}^n, \mathbf{x}_i)\}_{i=0}^n$ is a differentiable structure, and $\mathbb{R}P^n$ is a differentiable manifold of dimension n.

4 Differentiable maps between manifolds

Our next goal is to say when two manifolds are "the same" (isomorphic). For this we first need differentiable maps between manifolds. Differentiability of a map on M can only be tested through parametrizations, i.e. by going down to $\mathbb{R}^n$.

Definition 8. Let M_1^n and M_2^m be differentiable manifolds of dimensions n and m. A mapping $\varphi : M_1 \to M_2$ is *differentiable at* $p \in M_1$ if given a parametrization $\mathbf{y} : V \subset \mathbb{R}^m \to M_2$ at $\varphi(p)$ there exists a parametrization $\mathbf{x} : U \subset \mathbb{R}^n \to M_1$ at p such that $\varphi(\mathbf{x}(U)) \subset \mathbf{y}(V)$ and the mapping

$$\mathbf{y}^{-1} \circ \varphi \circ \mathbf{x} : U \subset \mathbb{R}^n \to \mathbb{R}^m$$

is differentiable at $\mathbf{x}^{-1}(p)$. A mapping which is differentiable at all points of M_1 is called *differentiable*.

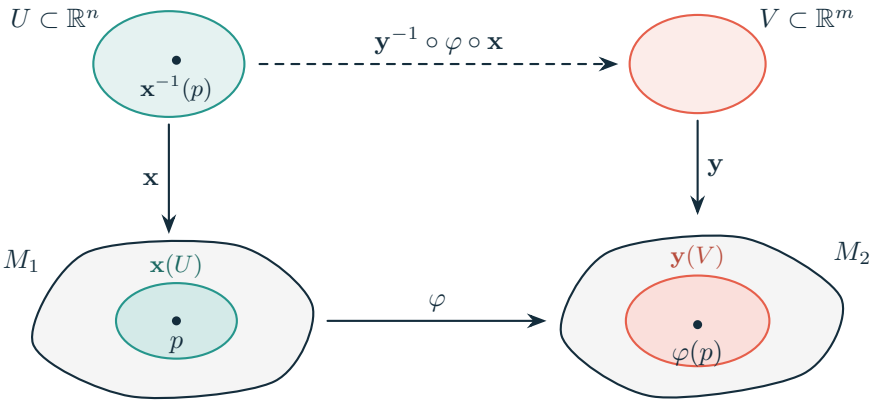


Figure 5: φ is differentiable at p if its expression in coordinates, $\mathbf{y}^{-1} \circ \varphi \circ \mathbf{x}$, is differentiable at $\mathbf{x}^{-1}(p)$. This is a map between open subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$, where we know what differentiable means.

The condition $\varphi(\mathbf{x}(U)) \subset \mathbf{y}(V)$ makes the composition $\mathbf{y}^{-1} \circ \varphi \circ \mathbf{x}$ defined on all of U. This is why $\mathbf{y}$ is chosen first and then U is taken small enough.

Question. Does differentiability of $\varphi : M_1 \to M_2$ depend on the choices made?

- It does *not* depend on the choice of parametrizations. If $\mathbf{x}', \mathbf{y}'$ are other parametrizations at p and $\varphi(p)$, then near $\mathbf{x}'^{-1}(p)$

  $$\mathbf{y}'^{-1} \circ \varphi \circ \mathbf{x}' = \left(\mathbf{y}'^{-1} \circ \mathbf{y}\right) \circ \left(\mathbf{y}^{-1} \circ \varphi \circ \mathbf{x}\right) \circ \left(\mathbf{x}^{-1} \circ \mathbf{x}'\right)$$

  and the outer two factors are changes of parameters, differentiable by condition (2). This is exactly why (2) is part of the definition.

- It *does* depend on the differentiable structures. For example, on $M = \mathbb{R}$ take the structure generated by $\mathbf{x}(t) = t$ and the one generated by $\mathbf{y}(t) = t^3$. Then $\mathbf{y}^{-1} \circ \mathrm{id} \circ \mathbf{x}(t) = t^{1/3}$ is not differentiable at 0, so $\mathrm{id} : (\mathbb{R}, \mathbf{x}) \to (\mathbb{R}, \mathbf{y})$ is not differentiable, and the two structures are different. (Still, $\varphi(t) = t^3$ from $(\mathbb{R}, \mathbf{x})$ to $(\mathbb{R}, \mathbf{y})$ is differentiable with differentiable inverse, since in coordinates both read $t \mapsto t$. So the two manifolds are "the same", which leads to the notion of isomorphism we are after.)